

Loops inside loops

Can the body of a loop have another loop within it? Yes! This is called a nested loop. It's like Russian dolls, where each doll fits inside a larger doll. In a nested loop, an inner loop runs inside an outer loop.



The loop variable of the outer loop is `hooray_counter`.

▷ One loop inside another

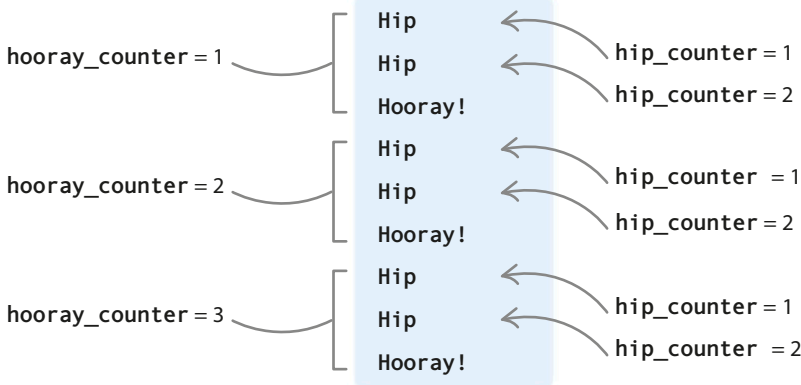
In this example, Emma has changed her “Keep Out” program into a “Three Cheers” program that prints “Hip, Hip, Hooray!” three times. Because each cheer includes the word “Hip” twice, she uses a nested loop to print it.

```
>>> for hooray_counter in range(1, 4):
      for hip_counter in range(1, 3):
          print('Hip')
          print('Hooray!')
```

The loop variable of the inner loop is `hip_counter`.

The body of the outer loop is indented 4 spaces.

The body of the inner loop is indented another 4 spaces.



EXPERT TIPS

Indent the body

The code in the body of a loop should be indented four spaces. If it isn't, Python will show an error message and the code won't run. With nested loops (one loop inside another), the body of the inner loop must be indented an extra four spaces. Python automatically indents new lines in loops, but you should always check that each line is indented by the correct number of spaces.



◁ How it works

The whole of the inner `for` loop is inside the body of the outer `for` loop. Each time we do one repeat of the outer loop, we have to do two repeats of the inner loop. This means the body of the outer loop is run three times in total, but the body of the inner loop is run six times.